

Bluestock Mutual Fund Analytics Capstone Project

Submitted By:

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Technologies:

Python | SQLite | Power BI | GitHub

Problem Statement & Objectives

Problem:

- Analyze mutual fund industry data and derive business insights.

Objectives:

- Build ETL Pipeline
- Perform EDA
- Compute Performance Metrics
- Conduct Advanced Analytics
- Develop Power BI Dashboard

Data Sources

Datasets Used:

- Fund Master
- NAV History
- AUM Data
- SIP Inflows
- Category Inflows
- Folio Counts
- Scheme Performance
- Investor Transactions
- Portfolio Holdings
- Benchmark Indices

Project Architecture

Raw CSV Files



Python ETL



Processed Data



SQLite Database



EDA & Analytics



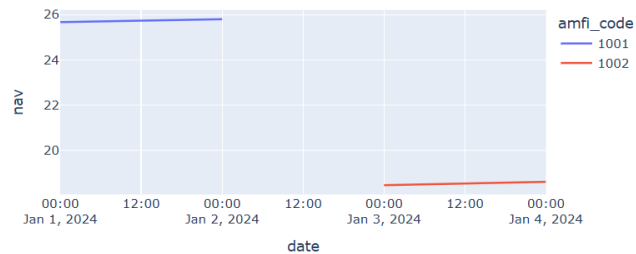
Power BI Dashboard



Business Insights

EDA Highlights (Part 1)

NAV Trend Analysis



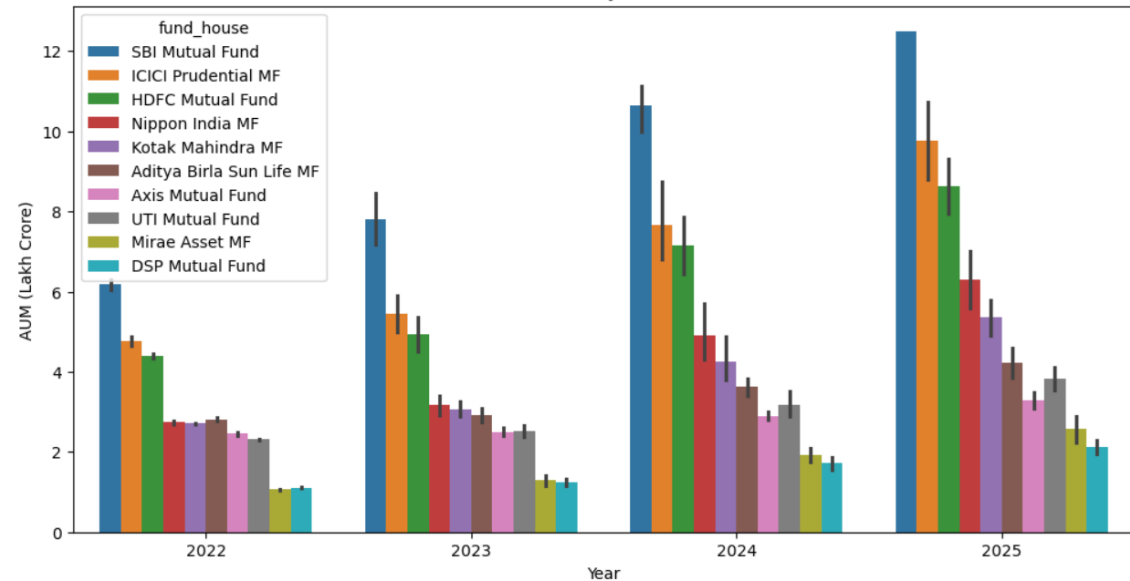
```
[4]: nav.head()
```

```
[4]:
```

	date	amfi_code	nav
0	2024-01-01	1001	25.67
1	2024-01-02	1001	25.80
2	2024-01-03	1002	18.45
3	2024-01-04	1002	18.60

```
plt.show()
```

AUM Growth by Fund House



- Mutual fund NAVs showed long-term growth.
- Industry AUM increased significantly from 2022–2025.

EDA Highlights (Part 2)

Monthly SIP Inflows (2022-2025)



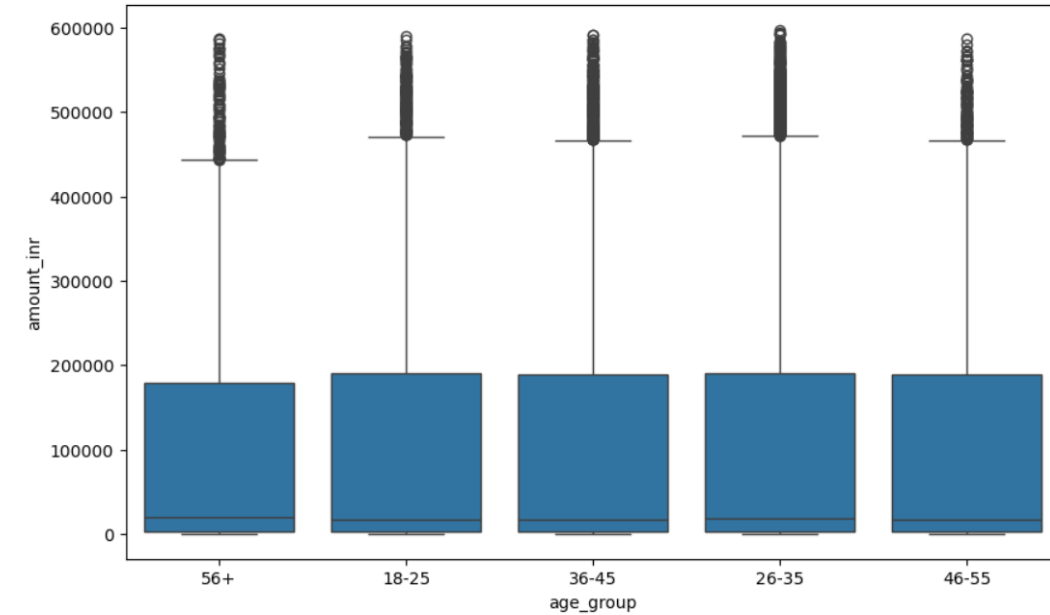
```
[25]: sip.head()
```

```
[25]:
```

	month	sip_inflow_crore	active_sip_accounts_crore	new_sip_accounts_lakh	sip_aum_lakh_crore	yoy_growth_pct
0	2022-01-01	11517	4.91	9.10	4.80	NaN
1	2022-02-01	11438	4.93	8.20	4.85	NaN
2	2022-03-01	12328	5.09	10.50	5.01	NaN
3	2022-04-01	11863	5.48	9.52	5.12	NaN
4	2022-05-01	12286	5.55	8.10	5.15	NaN

```
plt.show()
```

Investment Amount by Age Group



```
garden_counts = investment["garden"].value_counts()
```

- SIP inflows reached record levels.
- Younger investors showed strong participation.

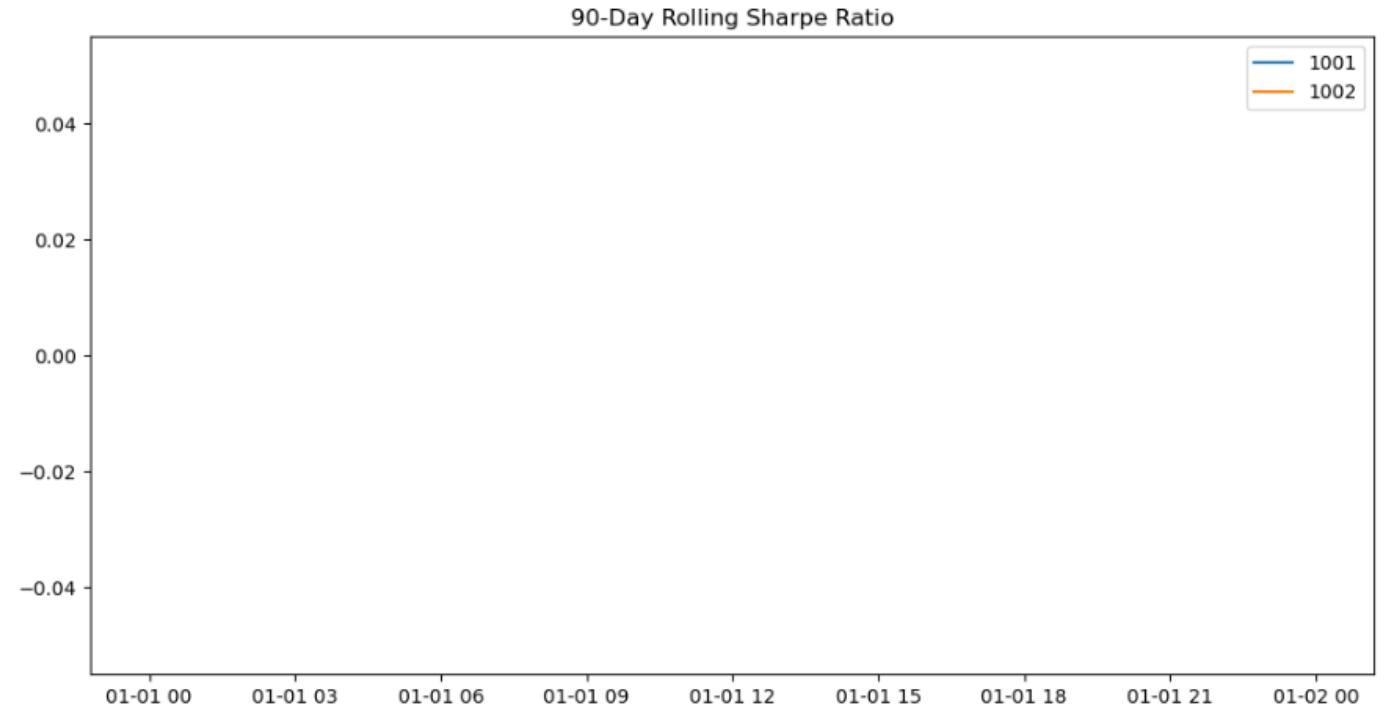
Performance Analytics

	scheme_name	fund_score
3	SBI Small Cap Fund - Direct Plan - Growth	72.75
22	Kotak Flexicap Fund - Regular - Growth	71.50
21	Kotak Emerging Equity Fund - Regular - Growth	70.50
29	ABSL Small Cap Fund - Regular - Growth	68.25
2	SBI Small Cap Fund - Regular Plan - Growth	64.38

- Top funds delivered superior risk-adjusted returns.

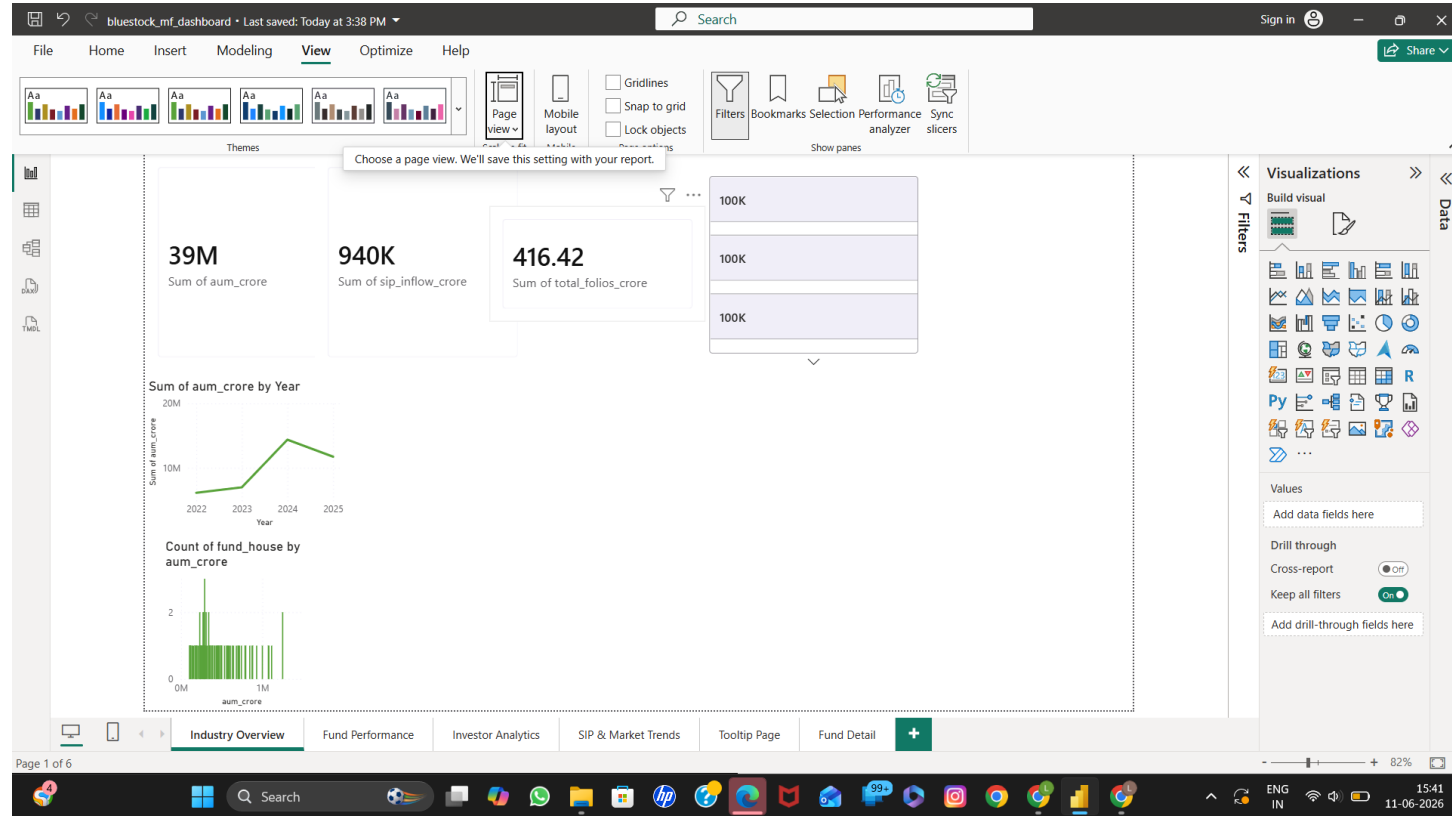
Advanced Analytics

	amfi_code	level_1	return
0	1001	VaR_95	0.005064
1	1001	CVaR_95	0.005064
2	1002	VaR_95	0.008130
3	1002	CVaR_95	0.008130

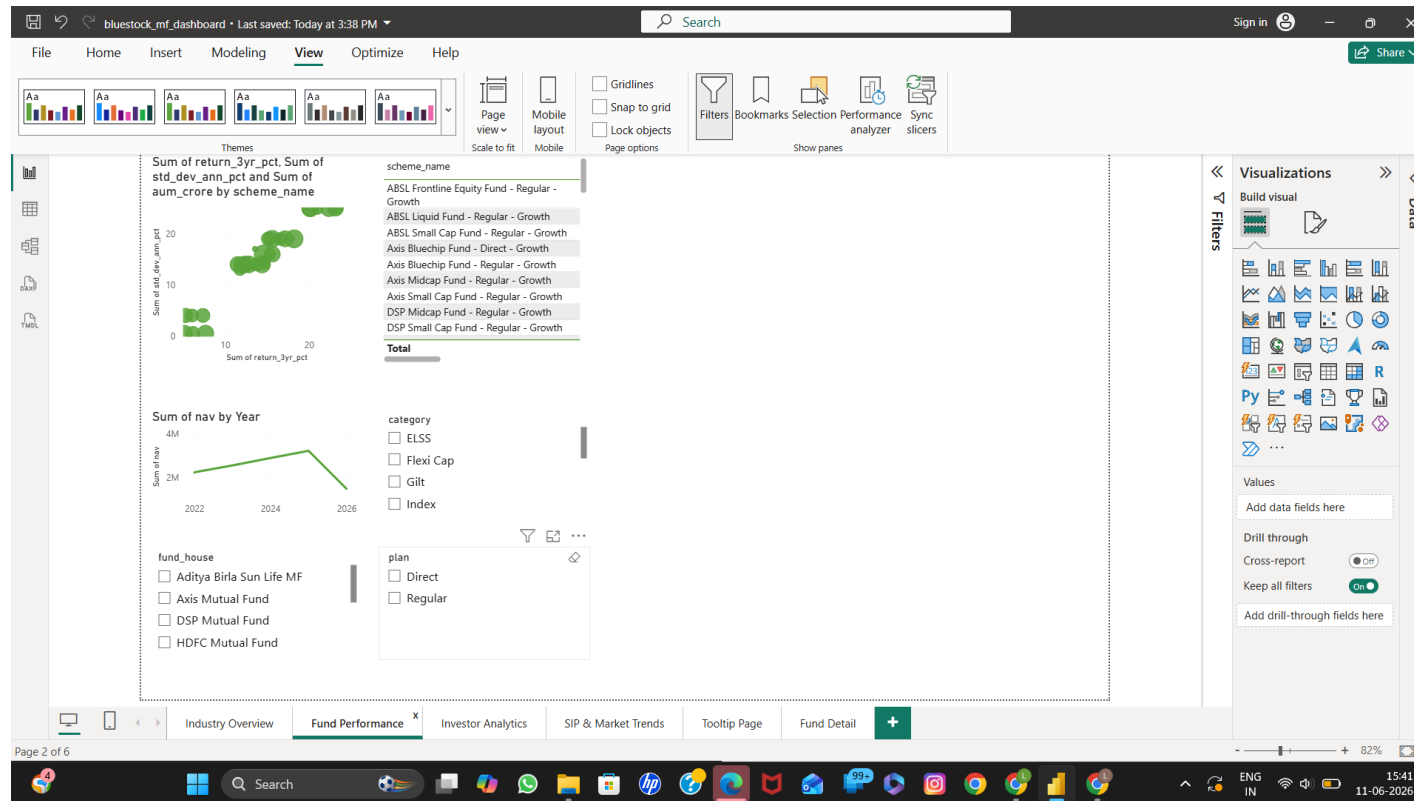


- Risk metrics identified high-volatility funds.
- Cohort analysis revealed long-term investor behavior.

Industry Overview



Fund Performance



Key Findings & Recommendations

Key Findings:

- SIP participation continues to grow.
- Equity funds dominate inflows.
- Top-ranked funds outperform benchmarks.

Recommendations:

- Integrate live market data.
- Add predictive analytics.
- Deploy dashboard online.

Thank you